Buck Converter Parameters Design Using Artificial Intelligence-based Genetic Algorithm

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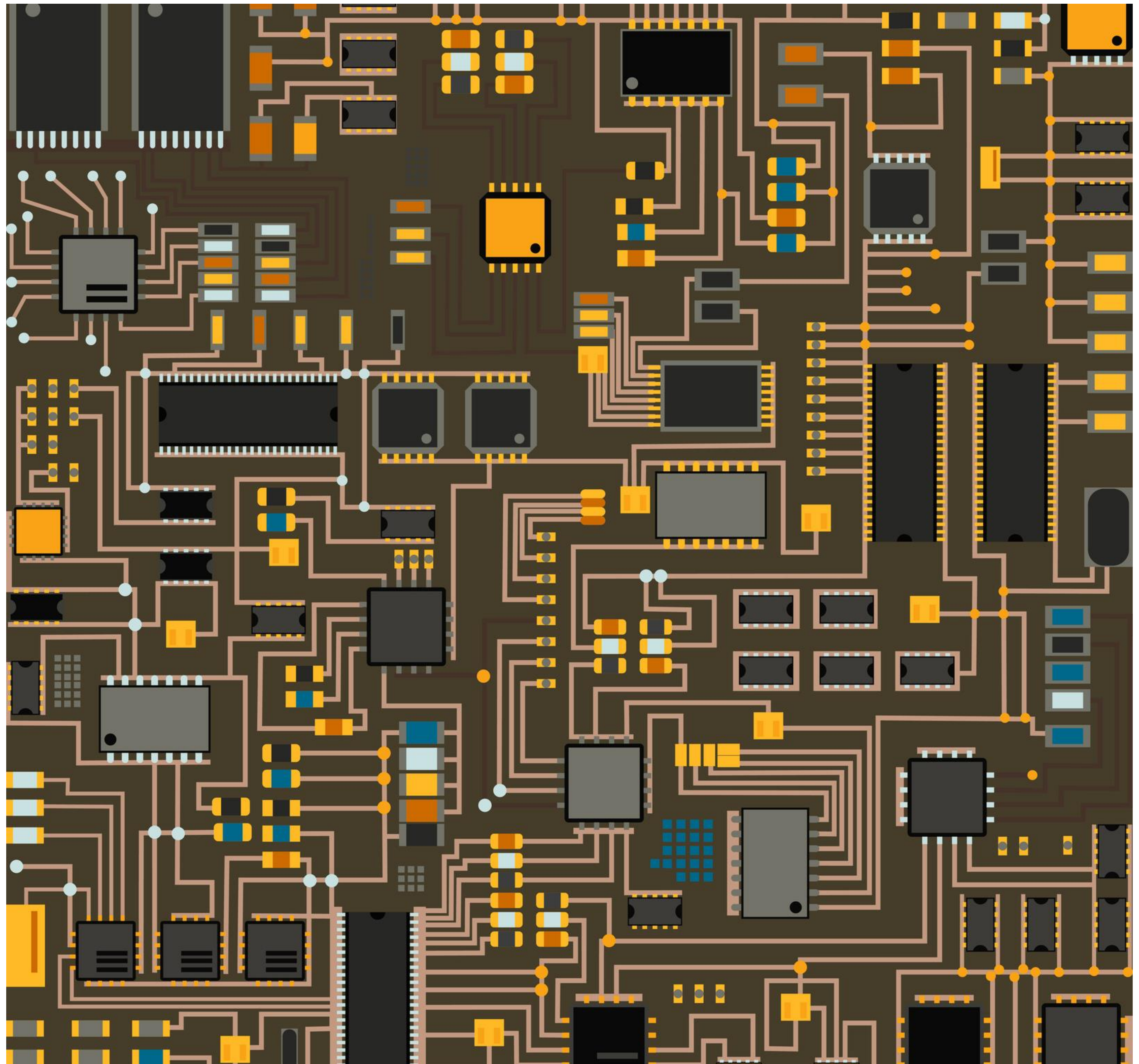
Outline

- Introduction
- AI-Based Circuit Design
- Conventional Circuit Design
- Results and Comparisons
- Conclusion



Introduction

Power Electronic → Buck Converter → Circuit Design

Introduction

- Circuit Design
  - Size
  - Cost
  - Efficiency
  - Reliability
  - Ripple
  - Transient Response
- Objective

Specification and Parameters Range

Parameter	Value
Input Voltage (V_{in})	48 V
Output Voltage (V_{out})	12 V
Output Power (P_o)	100 W
Current Ripple ($\Delta \mathrm{I_L}$)	$\leq 10\ \%$
Voltage Ripple (ΔV_{out})	$\leq 1\ \%$
Volume Limit (V_{lim})	$\leq 7\ \mathrm{cm}^3$
Target Efficiency	90 %

Parameter	Range of Values
Inductance (L)	30 μH – 2000 μH
Capacitance (C)	20 μF – 1000 μF
Switching Frequency (f_{sw})	20 kHz – 200 kHz
Dead Time (t_{dt})	175.2 ns – 262.8 ns
Duty Cycle (d_{cycle})	20 % – 35 %

AI-Based Circuit Design

PLECS $\xrightarrow{\text{Dataset}}$ Machine Learning $\xrightarrow{\text{Model}}$ Genetic Algorithm $\rightarrow$ Results

Conventional Circuit Design

$$D = \frac{V_{\text{out}}}{V_{\text{in(max)}} \times \eta}$$

$$t_{dead\_min} = \max(t_{d(off)} + t_f, t_{d(on)}) \times safety\,margin$$

$$t_{dead\_max} = t_{dead\_min} \times safety\,margin$$

$$C_{\text{out(min)}} = \frac{\Delta I_{\text{L}}}{8 \times f_{\text{sw}} \times \Delta V_{\text{out}}}$$

$$L = \frac{V_{\text{out}} \times (V_{\text{in}} - V_{\text{out}})}{\Delta I_{\text{L}} \times f_{\text{sw}} \times V_{\text{in}}} \times \eta$$

Conventional Method vs AI-Based

Parameter	Conventional Approach	AI-Based Approach
L	243.097 μH	285.8 μH
C	21.692 μF	127 μF
f_{sw}	40 kHz	39.841 kHz
t_{dt}	175.2 ns	180.90 ns
d_{cycle}	22.5 %	24 %

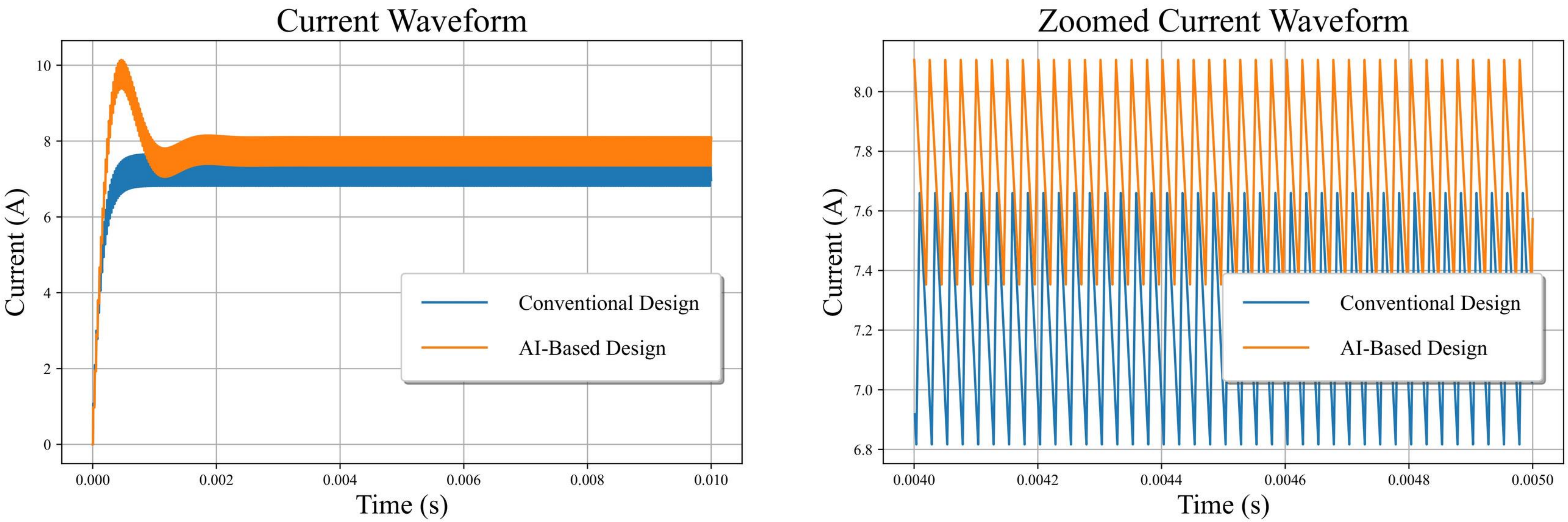
Result and Comparison

Parameter	Conventional Approach	AI-Based Approach
I_L (A)	7.228	7.719
ΔI_L (A)	0.843	0.753
V_{out} (V)	10.395	11.127
ΔV_{out} (V)	0.12	0.0187
Total Power Loss (W)	0.489	0.533
Volume (cm^3)	7.176	5.732
Efficiency (%)	99.354	99.383

Simulation Results

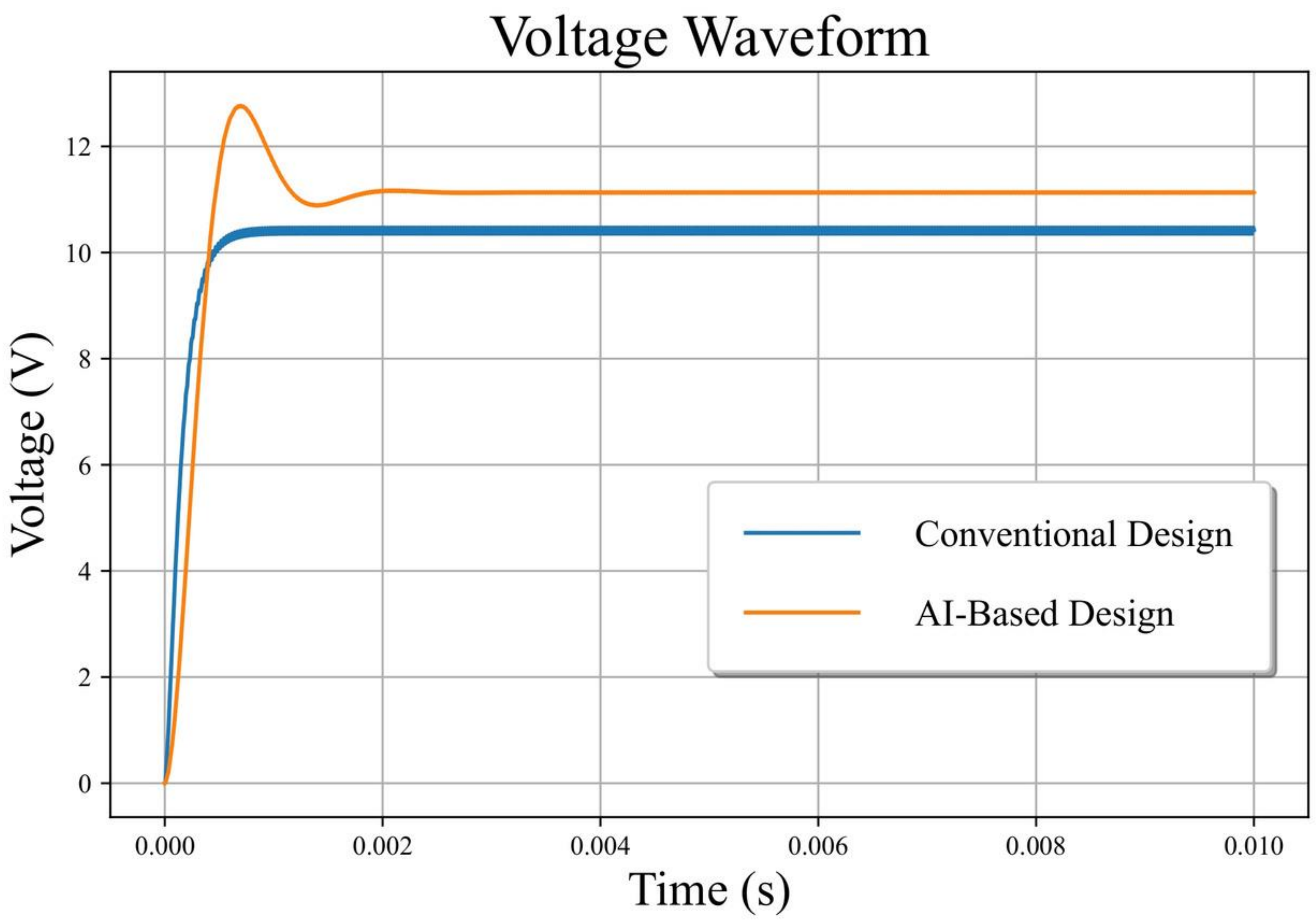
Result and Comparison

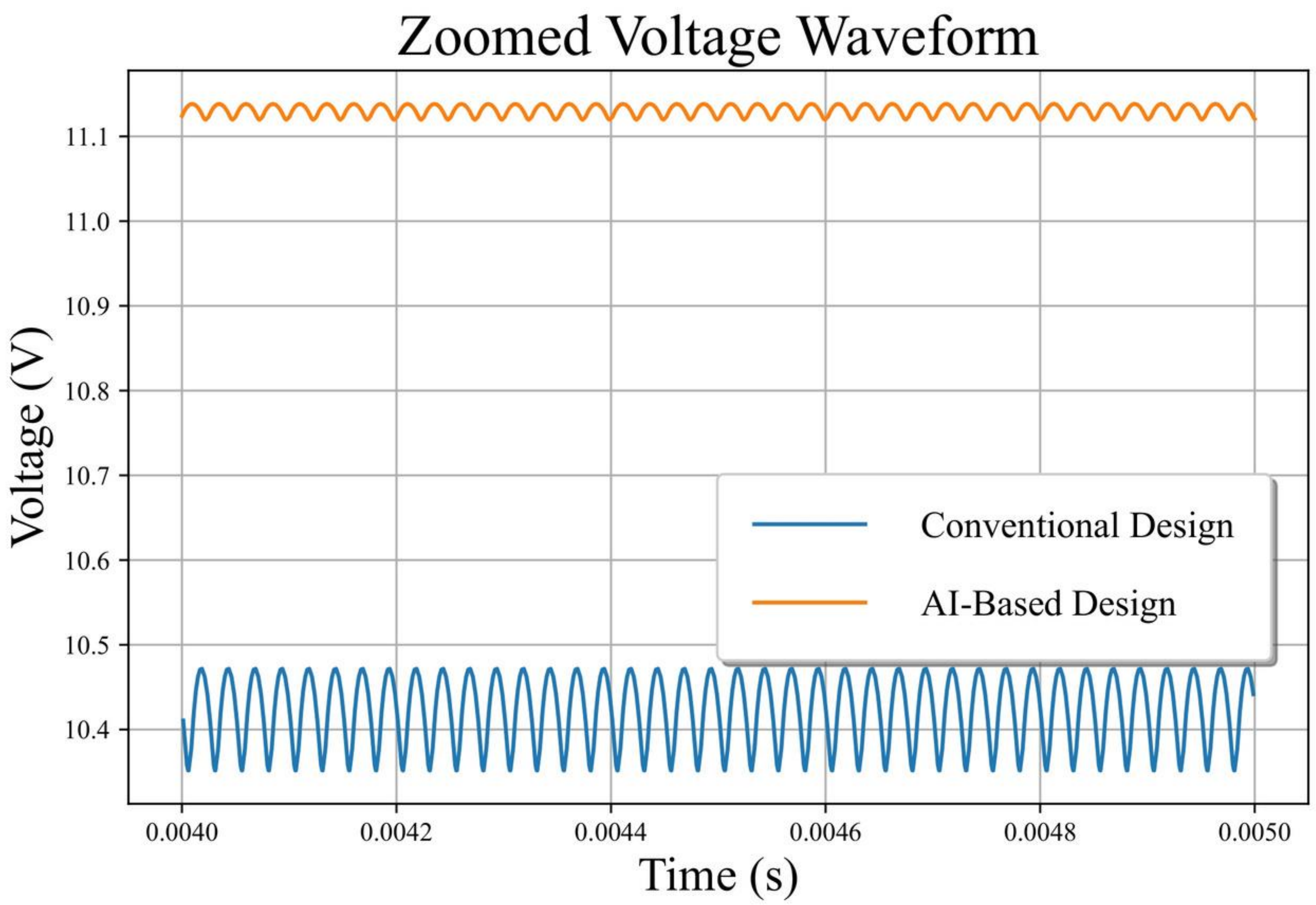
Current Waveform



Result and Comparison

Voltage Waveform





Conclusion

Efficiency

99.383% (AI-based) and 99.354% (Conventional)

Output Voltage Error

7.28% (AI-based) and 13.37% (Conventional)

Voltage Ripple

0.0187 V (AI-based) and 0.120 V (Conventional)

Current Ripple

0.753 A (AI-based) and 0.843 A (Conventional)

Volume Limit

5.732 cm^3 (AI-based) and 7.176 cm^3 (Conventional)

THANK YOU